VMWARECERTIFICATION

NSX-T

Data Center Network Virtualization & Security

Comprehensive Study Notes with Real-World Examples

PLATFORM	FOCUS AREA	EDITION	FORMAT
VMware NSX-T 3.x / 4.x	Network Virtualization	2026 Study Guide	Lora 12pt · PDF

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Chapter 1 — What Is VMware NSX-T?

VMware NSX-T (NSX Transformers) is a software-defined networking (SDN) and security platform. It creates a complete virtual network — switches, routers, firewalls, and load balancers — entirely in software, independent of the physical hardware underneath.

Think of NSX-T as the "VMware ESXi for networking". Just like ESXi virtualizes servers, NSX-T virtualizes the entire network. Physical switches and routers still exist, but NSX-T builds a programmable overlay network on top of them.

> ▯ Real-World Example — The Core Idea:
> A bank has 3 data centers. Without NSX-T, network teams manually configure physical switches and routers in each DC — taking weeks per change. With NSX-T, the entire network is defined in software. A policy change (like blocking a port or adding a firewall rule) is deployed in seconds across all 3 data centers from a single UI.

1.1 Why NSX-T? — Problems It Solves

Old Problem (Physical Networking)	How NSX-T Solves It	Business Impact
VLANs are limited to 4,094 IDs — not enough for large clouds and multi-tenant environments	VXLAN/Geneve tunnels support 16 million+ logical network IDs — virtually unlimited tenant isolation	Cloud providers can host thousands of customers on shared hardware, each with complete network isolation
Network changes require manual config on physical switches — takes days or weeks, prone to human error	All network config done in NSX Manager (UI or API) — changes deployed in seconds to all hosts	DevOps teams can provision networks in minutes alongside new application deployments
Security is perimeter-based — one firewall at the edge. Once inside, attackers move freely (east-west)	Distributed Firewall (DFW) runs inside every hypervisor kernel — filters every VM-to-VM packet	A compromised VM cannot spread to other VMs — stops ransomware lateral movement inside the data center
No visibility into VM-to-VM (east-west) traffic — blind spot for security and troubleshooting	NSX Intelligence and Traceflow give complete visibility into every packet path inside the virtual network	Security teams can see exactly what traffic each VM sends/receives and detect anomalies in real-time

Old Problem (Physical Networking)	How NSX-T Solves It	Business Impact
Moving a VM to another host breaks its IP address and VLAN config — network team must reconfigure	VM network policy travels with the VM — when VM migrates via vMotion, firewall rules and IP follow automatically	Zero network changes needed during VM migrations — enables true workload mobility across the data center

1.2 NSX-T vs NSX-V — What Changed?

Feature	NSX-V (Old)	NSX-T (Current)	Why It Matters
Platform Support	vSphere ONLY	vSphere, KVM, bare-metal, containers (K8s), public cloud	NSX-T works in hybrid/multi-cloud — not locked to VMware hypervisor
Management	Integrated with vCenter	Standalone NSX Manager — independent of vCenter	NSX-T can manage non-VMware environments
Control Plane	vSphere controller cluster (3 VMs)	NSX Manager cluster (3 VMs with embedded control plane)	Simplified architecture — fewer moving parts
Data Plane (Routing)	Kernel-based vSphere routing	N-VDS / Enhanced Data Path (EDPS) — higher throughput	Better performance for high-bandwidth workloads
Container Networking	Not supported	Native Kubernetes integration (NCP)	NSX-T is future-proof for cloud-native workloads
End of Life?	EOL announced — no new features	Active development — v4.x is latest	Migrate to NSX-T — NSX-V will not receive updates

▯ Note: VMware officially ended NSX-V general support. All new deployments and migrations should use NSX-T. The exam tests NSX-T concepts.

Chapter 2 — NSX-T Architecture Deep Dive

NSX-T uses a three-plane architecture: Management Plane, Control Plane, and Data Plane. Each plane has a specific job. Understanding which plane does what is critical for troubleshooting and the exam.

Plane	What It Is	Runs On	Main Job	Real Example
Management Plane	The brain — where admins configure everything. Stores desired state of the entire network.	NSX Manager VMs (3-node cluster for HA)	Accept admin input (UI/API), store config in database, push config to Control Plane	Admin creates a firewall rule in NSX Manager UI ▯ Management Plane stores it and pushes to Control Plane
Control Plane	The coordinator — calculates routing tables and distributes network state to data plane nodes.	Embedded inside NSX Manager cluster (same VMs in NSX-T 3.x+)	Compute MAC/IP tables, distribute logical topology info to all transport nodes	New VM powers on, gets MAC address ▯ Control Plane tells all transport nodes: "VM-X is at Host-3"
Data Plane	The worker — actually moves packets. Runs in the hypervisor kernel for wire-speed performance.	Each ESXi/KVM host (transport node) — runs as kernel module (N-VDS)	Forward packets, enforce firewall rules, apply QoS — all at line rate without Management Plane involvement	VM-A sends packet to VM-B ▯ Data Plane on Host-1 checks its local forwarding table and sends packet directly to Host-2 — no routing through a central device

▯ Real-World Example — Why 3-Plane Architecture Matters:
The Management Plane goes down (NSX Manager VM crashes). Traffic between VMs continues UNAFFECTED — the Data Plane has all the forwarding tables it needs. You just cannot make new configuration changes until Manager recovers. This is critical for production: management problems never cause network outages.

2.1 NSX Manager — Management Cluster

NSX Manager Detail	Specification / Explanation
Deployment Form	Virtual appliance (OVA) — runs as a VM on vSphere or KVM
Cluster Size	3-node cluster for production (Active-Active-Active). Single node allowed for lab only.
Embedded Services	All three: Management, Control Plane, and Policy API embedded in same 3 VMs
VIP (Virtual IP)	A single floating IP is configured — clients connect to VIP, it routes to the active Manager
Recommended RAM	24 GB RAM per node for medium deployments (up to 128 ESXi hosts)
Recommended vCPU	6–8 vCPUs per node
Storage	200 GB disk per node — stores NSX config database
Communication Port	Port 443 (HTTPS) for UI and API. Port 1235 for cluster sync between nodes.
Backup	NSX Manager supports scheduled backups to SFTP server — backup the config, not the VMs

▯ Note: NSX Manager itself needs network connectivity to all ESXi hosts (transport nodes). If the management network is broken, Manager cannot push config updates — but existing traffic keeps flowing on the data plane.

2.2 Transport Zones — The Foundation

A Transport Zone (TZ) defines the reach of an NSX-T overlay or VLAN network — which hosts can participate in that network. Think of it as a membership list for logical networking.

Transport Zone Type	How It Works	VLAN Required?	Use Case	Real Example
Overlay Transport Zone	Uses Geneve encapsulation to tunnel Layer 2 traffic over any IP network. Creates logical (virtual) networks independent of physical VLANs.	No — tunnels over any IP network. Only needs IP reachability between transport nodes.	VM-to-VM co mmunication across hosts, micro-segme ntation, logical routing	Overlay TZ "Production-Overlay" — all 20 ESXi hosts are members. Any Segment on this TZ can span all 20 hosts without VLAN coordination with the physical network team.
VLAN Transport Zone	Bridges NSX-T logical networks to physical VLAN-based networks. No encapsulation — traffic is tagged with 802.1q VLAN IDs.	YES — each VLAN TZ Segment maps to a specific VLAN ID on the physical switch	Connecting NSX-T workloads to physical (non-virtualized) servers and external network equipment	VLAN TZ "Physical-VLAN-TZ" — Segment "VLAN-100-Storage" bridges NSX-T VMs to physical storage array on VLAN 100

▯ Real-World Example — Transport Zone Design:
Company has 2 clusters: Production (16 hosts) and DMZ (4 hosts). They create 2 Overlay TZs: "Prod-Overlay-TZ" (16 hosts) and "DMZ-Overlay-TZ" (4 hosts). This ensures DMZ workloads cannot accidentally reach Production segments even if misconfigured — they are in completely separate transport zones.

2.3 Transport Nodes — NSX-T Participants

A Transport Node is any host that participates in NSX-T networking — ESXi hosts, KVM hosts, bare-metal servers, or NSX Edge nodes. Transport Nodes run the N-VDS (NSX Virtual Distributed Switch) and the data plane components.

Transport Node Type	What It Is	N-VDS Role	Key Requirement
ESXi Host Transport Node	vSphere host with NSX-T kernel modules installed	Hosts VM workloads — forwards east-west VM traffic at hypervisor kernel speed	NSX-T kernel VIBs installed, TEP (Tunnel Endpoint) VMkernel port configured
KVM Host Transport Node	Linux KVM host with NSX-T OVS kernel module	Same as ESXi — hosts VMs and containers	Open vSwitch (OVS) + NSX-T agent installed, TEP port configured
Bare-Metal Transport Node	Physical server with NSX-T agent (no hypervisor)	Applies NSX-T security policies directly to physical workloads	NSX-T bare-metal agent installed on Linux/Windows OS
NSX Edge Transport Node	Dedicated VM or bare-metal appliance for north-south routing, NAT, VPN, load balancing	Connects NSX-T overlay to the physical network — uplink between virtual and physical world	Physical NIC uplinks, TEP port, BGP/OSPF config for physical network connectivity

Chapter 3 — Logical Switching & Segments

In NSX-T, traditional VLANs are replaced by Segments (also called Logical Switches). A Segment is a virtual Layer 2 broadcast domain that can span multiple physical hosts without any physical switch configuration.

3.1 How Geneve Encapsulation Works

NSX-T Overlay networks use Geneve (Generic Network Virtualization Encapsulation) to tunnel Layer 2 frames over an IP network. This is how VMs on different hosts can be on the same logical L2 network without any VLAN on the physical switch.

Step	What Happens	Where It Happens
1 — VM Sends Frame	VM-A sends an Ethernet frame to VM-B. Frame has VM-A MAC as source, VM-B MAC as destination. Standard Layer 2 frame.	Inside the VM — normal networking
2 — N-VDS Intercepts	The N-VDS (NSX Virtual Distributed Switch) on Host-1 intercepts the frame before it hits the physical NIC.	ESXi/KVM kernel — data plane
3 — Geneve Encapsulation	N-VDS wraps the original Ethernet frame inside a new UDP packet (Geneve header). Outer IP header uses the TEP (Tunnel Endpoint) IPs of the source and destination hosts. Geneve header contains the VNI (Virtual Network Identifier — up to 16 million values).	ESXi/KVM kernel — N-VDS TEP
4 — Physical Network Carries It	The encapsulated UDP packet travels across the physical network (switches/routers) using standard IP routing. Physical switches see only normal IP packets — they have no knowledge of the virtual network inside.	Physical network infrastructure
5 — Decapsulation on Destination	N-VDS on Host-2 receives the UDP packet, strips the Geneve header, and delivers the original Ethernet frame to VM-B.	ESXi/KVM kernel — N-VDS TEP on Host-2

▯ *Exam Tip: Geneve uses UDP port 6081. Physical switches/firewalls must allow UDP 6081 between all transport node TEP IPs. Blocking this port breaks ALL NSX-T overlay networking. This is a very common production mistake!*

3.2 Segments — Virtual Networks

Segment Property	Details	Example Value
Segment Type	Overlay (uses Geneve/VXLAN tunnel) or VLAN (maps to physical VLAN)	Overlay for workload segments, VLAN for uplink segments
VNI (Virtual Network ID)	Unique 24-bit ID assigned to each overlay segment — like a VLAN ID but with 16 million possible values	Segment "Web-Tier" gets VNI 100001, "DB-Tier" gets VNI 100002
Transport Zone	Which TZ this segment belongs to — determines which hosts can connect VMs to this segment	Segment "Production-Web" in "Prod-Overlay-TZ" — only hosts in that TZ can have VMs on this segment
Gateway (Uplink)	Optional — connect the segment to a Tier-1 or Tier-0 Gateway for routing	Segment "Web-Tier" connected to T1-Gateway "T1-Prod" for inter-segment routing
Segment Profiles	Additional policies: MAC discovery, IP discovery, security profiles, QoS	IP Discovery Profile ensures ARP is suppressed — reduces broadcast flooding in large segments

▯ Real-World Example — Segment Creation:
E-commerce company needs 3 tiers: Web, App, and Database. NSX admin creates 3 Overlay Segments: "Seg-Web" (VNI 100010), "Seg-App" (VNI 100020), "Seg-DB" (VNI 100030). Each tier is isolated at L2. Web VMs connect to Seg-Web, App VMs to Seg-App, DB VMs to Seg-DB. Traffic between tiers must go through the Tier-1 Gateway — where firewall rules can be applied.

3.3 ARP Suppression — Why It Matters

In a traditional VLAN, an ARP request (Who has IP x.x.x.x?) is broadcast to every device on the segment. In a large data center with thousands of VMs, this creates massive broadcast traffic (ARP storms).

NSX-T solves this with ARP Suppression: NSX Manager maintains an IP-to-MAC mapping table for all VMs. When a VM sends an ARP request, the N-VDS on the local host answers it directly from its local table — the ARP never hits the physical network as a broadcast.

> □ Real-World Example — ARP Suppression in Action:
> 5,000 VMs on one segment. VM-1 needs MAC address for VM-2's IP. Without ARP Suppression: broadcast to all 5,000 VMs. With NSX-T ARP Suppression: N-VDS on VM-1's host checks its local IP-MAC table, finds VM-2's MAC, replies instantly. Zero broadcast traffic — 4,999 VMs never receive the ARP. Massively reduces network overhead in large environments.

Chapter 4 — Logical Routing: Tier-0 and Tier-1 Gateways

NSX-T replaces physical routers with Logical Routers implemented entirely in software. There are two types: Tier-0 (T0) — the north-south gateway to the physical world, and Tier-1 (T1) — the tenant/application router connecting segments to the T0.

4.1 Tier-0 Gateway — The Border Router

The Tier-0 Gateway is the edge of the NSX-T network. It connects the overlay virtual network to the physical network (data center core switches, internet routers). It supports dynamic routing protocols (BGP, OSPF) to exchange routes with physical routers.

T0 Feature	Details	Real-World Analogy
North-South Routing	Routes traffic between the NSX-T overlay (virtual) and the physical network (external)	Like the lobby of a building — all traffic entering or leaving must pass through here
BGP/OSPF Support	Peering with physical TOR (Top-of-Rack) switches using BGP or OSPF to exchange routes dynamically	T0 tells physical switches "I have subnet 192.168.100.0/24 behind me" — physical switch adds this route automatically
ECMP (Equal-Cost Multi-Path)	Traffic load balanced across multiple T0 SR (Service Router) instances for high throughput and redundancy	Like having 4 checkout lanes at a supermarket — load is spread evenly, one lane closing does not stop everyone
NAT (Network Address Translation)	Translates private VM IPs to public IPs for internet access (SNAT) or inbound connections (DNAT)	SNAT: 100 VMs all appear to come from one public IP. DNAT: public IP:80 redirected to internal web server VM.
VPN (IPSec/L2VPN)	Site-to-site IPSec VPN to branch offices or cloud providers. L2VPN extends Layer 2 segments to remote sites.	Branch office in Mumbai connects to data center in Delhi over encrypted IPSec tunnel — same logical network

T0 Feature	Details	Real-World Analogy
Failover Mode	Active-Standby (one T0 active, one on standby) or Active-Active (both process traffic via ECMP)	Hospital uses Active-Standby T0 — zero packet loss during T0 VM failure because standby takes over in < 1 second

▯ Real-World Example — T0 BGP Peering:
Data center has two Top-of-Rack (TOR) switches. NSX-T T0 Gateway runs as Active-Active with 2 Edge VMs. Each Edge VM peers with both TOR switches via eBGP. T0 advertises overlay subnet 10.10.0.0/16 to both TOR switches. If one Edge VM fails, the other continues BGP sessions and traffic keeps flowing. Physical switches automatically re-route through the surviving Edge VM.

4.2 Tier-1 Gateway — The Tenant Router

The Tier-1 Gateway provides routing between multiple Segments connected to it, and then connects up to a T0 for north-south access. Each application or business unit typically gets its own T1.

T1 Feature	Details	Example
Inter-Segment Routing	Routes between Segments connected to the same T1 — without traffic leaving the hypervisor host (Distributed Routing)	Web VM (Seg-Web, 10.10.1.0/24) pings DB VM (Seg-DB, 10.10.3.0/24) — routed inside Host-1 kernel. Zero physical network hops.
Uplink to T0	A single link connects T1 to T0. T1 advertises its subnets up to T0 automatically.	T1 "T1-EcommerceApp" connected to T0 "T0-DC-Edge". T0 sees all T1 subnets and advertises them to physical network.
Service Insertion	T1 can apply L4-L7 services: NAT, Load Balancer, DNS Forwarder	T1 has a Load Balancer configured: VIP 10.10.1.100 distributes to 5 web server VMs behind it.
Centralized vs Distributed Services	Some T1 services run distributed (in every host kernel) — fast. Some run centralized (on Edge VM only) — for stateful services.	T1 Distributed Router: routes packets inside hypervisor at near line-rate. T1 Service Router (on Edge): handles NAT, LB where state must be tracked.

T1 Feature	Details	Example
Multiple T1s per T0	One T0 can have dozens of T1 gateways connected — used in multi-tenant or multi-application environments	MSP with 50 customers: each customer gets a dedicated T1. All 50 T1s connect to one shared T0. Full isolation between tenants.

4.3 Routing Architecture: Distributed vs Centralized

Routing Type	Where It Runs	Performance	What Traffic It Handles	Example
Distributed Router (DR)	Runs inside EVERY ESXi/KVM host kernel as part of N-VDS	Line-rate — handled inside hypervisor, no external hop needed	East-west traffic: VM-to-VM routing across different subnets on the same or different hosts	Web server (10.10.1.5) sends request to App server (10.10.2.5) — DR on the same host routes it instantly without leaving the host
Service Router (SR)	Runs only on NSX Edge VMs — centralized service point	Good but requires traffic to exit the host and reach Edge VM	North-south traffic, stateful NAT, VPN termination, Load Balancing, BGP peering	Web server needs to reach internet (8.8.8.8) — traffic goes to DR on host, then to SR on Edge VM, then out through T0 to physical network

▯ Real-World Example — Distributed Routing in Action:
Host-1 has 2 VMs: VM-A on subnet 192.168.1.0/24 and VM-B on subnet 192.168.2.0/24. Both subnets are connected to the same T1 Gateway. VM-A pings VM-B. The Distributed Router running inside Host-1's kernel routes the packet directly from VM-A to VM-B — the packet NEVER leaves Host-1. No latency from going through a physical router or even an NSX Edge VM. This is the performance advantage of NSX-T distributed routing.

Chapter 5 — Distributed Firewall (DFW) & Security

The Distributed Firewall (DFW) is NSX-T's most powerful security feature. Unlike a traditional firewall (a physical box at the network edge), the DFW runs inside every hypervisor kernel — right at the virtual NIC (vNIC) of each VM. Every single packet sent or received by every VM is inspected by the DFW.

5.1 Why DFW Is Revolutionary

Traditional Perimeter Firewall	NSX-T Distributed Firewall (DFW)
Single physical box at the network edge — East-West (VM-to-VM) traffic is NOT inspected	Runs in every hypervisor kernel — inspects EVERY packet including East-West VM-to-VM traffic
Bottleneck — all traffic must physically go through one device	No bottleneck — each host processes its own traffic independently. Scales linearly with hosts.
Firewall rules based on IP addresses — VMs change IPs, rules break	Rules based on VM tags and security groups — dynamic. VM moves or changes IP, rules follow automatically.
Adding a new firewall rule takes days — change management, physical device access	New rule deployed via API or UI in seconds — instant enforcement across all hosts simultaneously
No security between VMs in the same VLAN — lateral movement is easy for attackers	Micro-segmentation: each VM can have its own firewall policy — lateral movement is blocked at the vNIC level

> ▯ Real-World Example — Ransomware Containment:
> Ransomware infects an accounting VM (10.10.5.50). Without DFW: ransomware spreads freely to all other VMs on the same VLAN — potentially encrypting hundreds of systems. With NSX-T DFW: micro-segmentation rules only allow accounting VMs to talk to the finance database on port 1433. Ransomware tries to connect to other systems — DFW blocks it at the vNIC. Infection is contained to one VM.

5.2 DFW Policy Structure

DFW rules are organized in a hierarchy. Understanding this hierarchy is critical for correct rule processing.

Level	Object	Priority	Purpose	Example
1 (Highest)	Ethernet Rules	Processed first	L2 rules — block/allow based on MAC addresses	Block unknown MAC addresses from sending traffic
2	Emergency Rules (Quarantine)	Very high priority	Quickly isolate a compromised VM — override all other rules	Security team detects breach: one click quarantines VM-X, blocking all its traffic immediately
3	Infrastructure Rules	High priority	Allow essential services (DNS, NTP, vCenter, backup agents) that must always work	Allow UDP 53 from all VMs to DNS servers — no application team can accidentally block DNS
4	Environment Rules (Isolation)	Medium-high	Separate major environments: Production, Dev, Test, DMZ cannot talk to each other	Block all traffic from Production VMs to Dev VMs — dev environment cannot reach prod database
5	Application Rules	Medium	Application-specific allow rules — the main policy	Allow TCP 8080 from Web-Security-Group to App-Security-Group. Allow TCP 1433 from App to DB.
6 (Lowest)	Default Rule	Lowest — catch-all	Applied if no above rule matches. Best practice: Default Deny (block everything not explicitly allowed)	Any traffic not matching rules 1-5 is dropped and logged — zero trust model

5.3 Security Groups & Tags — Dynamic Firewalling

Security Groups are collections of workloads (VMs, segments, containers) that share the same security policy. Tags are labels applied to VMs — Security Groups can be defined to automatically include all VMs with specific tags.

Concept	How It Works	Dynamic?	Example
Static Security Group	Admin manually adds specific VMs to a group	No — must be manually updated when VMs are added /removed	SG-WebServers contains: vm-web-01, vm-web-02, vm-web-03
Dynamic Security Group (Tag-based)	Group automatically includes any VM with matching tags	YES — VM gains tag ▯ instantly added to group and gets all policies	SG-Production includes all VMs tagged "Env:Production". New VM gets tag ▯ DFW rules apply instantly.
IP-based Group	Group defined by IP address or subnet range	Semi — works for fixed IPs but breaks if VMs change IP	SG-DMZ = subnet 10.10.0.0/24
Segment-based Group	All VMs connected to a specific NSX-T Segment	Yes — auto matically includes any VM on that segment	SG-Finance = all VMs on Segment "Finance-Net"

> ▯ Real-World Example — Dynamic Security Policy:
> Cloud infrastructure team has an automated VM deployment pipeline. When a new web server VM is deployed, automation scripts apply the NSX-T tag "Tier:Web". The DFW Security Group "SG-WebTier" is defined as "all VMs where Tag = Tier:Web". The moment the tag is applied, DFW rules automatically allow HTTPS on port 443 from the internet and deny all other inbound connections — all without a human touching the firewall config.

5.4 Identity Firewall (IDFW)

IDFW extends DFW to apply firewall rules based on Active Directory user identity — not just VM IP addresses. When a user logs into a VM, NSX-T identifies them and applies user-specific firewall rules.

> □ Real-World Example — IDFW Use Case:
> Finance department: When CFO logs into any Windows VM, IDFW automatically applies a policy allowing access to the finance application on port 8443 and SAP on port 3200. When a developer logs into the same VM later, they get a completely different policy — only access to dev servers. The VM IP never changes, but the rules change based on WHO is logged in.

Chapter 6 — NSX Edge Nodes & Gateway Services

The NSX Edge Node is a dedicated VM (or bare-metal appliance) that provides centralized gateway services — services that require a single state point, like NAT, VPN, and Load Balancing. Edge Nodes are the bridge between NSX-T's virtual overlay and the physical network.

6.1 NSX Edge Node Specs

Edge Form Factor	vCPU	RAM	Max Throughput	Supported Services	Use When
Small (Lab only)	2	4 GB	Not for production	Basic lab testing only	Proof of concept only — never production
Medium	4	8 GB	~4 Gbps	Routing, basic NAT, small VPN	Branch offices, small environments < 10 VMs
Large	8	32 GB	~8 Gbps	All services including Load Balancer	Most production deployments
Bare Metal	36+ cores	256 GB	100+ Gbps	Full services + SR-IOV for line-rate	Telcos, network service providers, hyperscale

▫ *Note: Always deploy Edge Nodes in pairs (minimum 2) for high availability. Edge Nodes are placed in an Edge Cluster — T0/T1 Service Routers fail over between Edge Nodes in the cluster.*

6.2 NAT — Network Address Translation

NAT Type	Direction	How It Works	Real-World Example
SNAT (Source NAT)	Outbound — VM to external network	Changes source IP of VM packets to a public IP before sending to internet. Multiple VMs share one public IP.	100 internal VMs (10.10.x.x) all appear to internet as single public IP 203.0.113.50. Internet sees only one IP regardless of which internal VM initiates the connection.
DNAT (Destination NAT)	Inbound — external to VM	Maps an external public IP:port to an internal VM IP:port. External clients connect to public IP, NAT redirects to internal VM.	External users hit https://company.com (203.0.113.50:443). DNAT rule translates this to internal web server VM 10.10.1.10:443. Web server never needs a public IP.
Reflexive NAT (No-DNAT)	Both directions	No translation applied — used to exclude specific traffic from NAT when other NAT rules would otherwise match	Management traffic from monitoring server 10.10.99.5 should NOT be NATed — add Reflexive NAT rule that matches before SNAT, keeps original source IP intact.

6.3 Load Balancer (NSX-T LB)

NSX-T's built-in Load Balancer distributes incoming traffic across multiple backend server VMs. Runs on the T1 Service Router (on NSX Edge VM).

LB Concept	Explanation	Example
Virtual Server (VIP)	The IP:port that clients connect to. NSX-T LB receives traffic here.	VIP: 10.10.1.100:443 — all HTTPS traffic for the app goes to this one address
Server Pool	The group of backend VM IPs that traffic is distributed to.	Pool: [10.10.1.11:443, 10.10.1.12:443, 10.10.1.13:443] — 3 web server VMs
Load Balancing Algorithm	Round Robin: each connection goes to next server in order. Least Connection: goes to server with fewest active sessions. IP Hash: same client always goes to same server.	Shopping cart application uses IP Hash — ensures user's session always reaches the same server (no session sharing needed)

LB Concept	Explanation	Example
Health Monitor	LB regularly tests each backend — if one fails, traffic is automatically redirected to healthy servers only.	Health check: HTTP GET /health every 5 seconds. If web-server-02 returns HTTP 500, LB removes it from pool instantly. No manual intervention.
Persistence (Session Stickiness)	Ensures all requests from one client go to the same backend VM — needed for stateful applications.	User is browsing an e-commerce site. Cookie persistence ensures all their clicks go to web-server-01 — cart data stays consistent.

▯ Real-World Example — NSX-T Load Balancer Setup:
HR application runs on 4 backend VMs. NSX-T LB VIP = 10.10.50.100:8080. Server pool = hr-app-vm-01 to vm-04. Algorithm = Least Connections. Health Monitor: TCP port 8080 every 10 seconds. During lunch hour peak: LB automatically distributes 400 concurrent users across all 4 VMs. hr-app-vm-03 crashes — health check fails in 10 seconds, LB stops sending traffic to it. Users experience zero interruption, remaining 3 VMs handle the load.

6.4 VPN Services

VPN Type	Use Case	Protocol	Example
IPSec Site-to-Site VPN	Connect two physical data centers or a data center to cloud (AWS/Azure) over an encrypted tunnel	IKEv1/IKEv2, ESP encryption	Delhi HQ and Mumbai branch: IPSec tunnel allows branch users to access data center apps as if on local network. Traffic is AES-256 encrypted in transit.
L2VPN (Layer 2 Extension)	Extend an NSX-T Segment across WAN to a remote site — remote VMs appear to be on the same L2 subnet	L2 over IPSec	Data center migration: Old DC (Mumbai) and New DC (Pune). L2VPN extends the production VLAN. VMs migrate to new DC with ZERO IP address change — no network reconfiguration needed.

VPN Type	Use Case	Protocol	Example
Remote Access VPN (SSL-VPN)	Individual remote workers connect to corporate network over encrypted VPN tunnel	SSL/TLS (client-based)	Work-from-home employee runs NSX VPN client — gets full network access to internal resources as if in the office. Multi-factor auth supported.

Chapter 7 — Micro-Segmentation & Zero Trust

Micro-segmentation is the practice of applying individual security policies to each workload — isolating VMs from each other even if they are on the same network segment. It is NSX-T's answer to the "flat network" security problem.

7.1 Zero Trust Network Model

"Never trust, always verify" — the Zero Trust security model assumes that threats exist both outside AND inside the network. Every connection, regardless of source, must be authenticated and authorized.

Zero Trust Principle	What It Means in NSX-T	Real Example
Default Deny Everything	DFW default rule = Block All. Only explicitly allowed connections are permitted.	New VM powers on — it can talk to nobody until a security admin creates an allow rule for it. Zero implicit trust.
Least Privilege Access	Each workload can only communicate with exactly what it needs — nothing more	Web server allowed: inbound TCP 443 from internet, outbound TCP 8080 to app server. Cannot reach database, DNS, or any other system.
Identity-Based Access	Rules based on WHO/WHAT the workload is (tags, AD identity) — not where it is (IP address)	Tag "DB:Oracle" on VMs automatically applies Oracle DB security policy — regardless of which host or IP the VM runs on
Continuous Verification	NSX Intelligence continuously monitors traffic patterns — anomalies trigger alerts	VM suddenly starts making connections to 500 different IPs (port scanning behavior) — NSX Intelligence alerts SOC team in real-time
Breach Containment	Even if one VM is compromised, DFW contains the breach to that VM — blast radius is one VM	Compromised VM tries to attack others — DFW blocks all unauthorized connections at the vNIC level. Attacker is trapped.

7.2 Designing Micro-Segmentation

A typical micro-segmentation design uses a 5-tier model:

Tier	Zone Name	Workloads	Allowed Inbound	Allowed Outbound
Tier 1	DMZ	Public-facing web servers, reverse proxies	HTTPS 443 from Internet	TCP 8080 to App Tier only
Tier 2	Application	App servers, API gateways, middleware	TCP 8080 from DMZ only	TCP 1433/3306 to DB Tier, TCP 443 to External APIs
Tier 3	Database	SQL, Oracle, NoSQL databases	TCP 1433/3306 from App Tier only	TCP 443 to Backup Server
Tier 4	Management	vCenter, NSX Manager, monitoring, backup	HTTPS 443 from Admin IPs only, ICMP from monitoring	All internal (management systems need broad access for their job)
Tier 5	Quarantine	Suspected compromised VMs	NOTHING (complete isolation)	NOTHING — quarantined VM has zero network access

▯ Real-World Example — Micro-Segmentation for PCI DSS Compliance:
E-commerce company must isolate its payment processing VMs for PCI DSS compliance. NSX admin creates Security Group "SG-PCI-Scope" with tag "Compliance:PCI". DFW rules: ONLY allow payment processor IPs to reach PCI VMs on TCP 443. All other traffic to/from PCI VMs = blocked. Traditional approach: dedicate an entire VLAN + physical firewall = expensive. NSX-T approach: PCI VMs can be on shared hardware, micro-segmentation provides the isolation = 70% cost reduction.

Chapter 8 — Operations, Monitoring & Troubleshooting

8.1 Traceflow — Packet-Level Troubleshooting

Traceflow is NSX-T's built-in packet tracing tool. It injects a synthetic (test) packet into the network and shows you exactly where it goes, what rules it hits, and where it is dropped — without sending real traffic.

Traceflow Capability	Details	Example Use
Unicast Trace	Trace a packet from one specific VM to another VM (or external IP)	Test: Can vm-web-01 reach vm-db-01 on TCP 1433? Traceflow shows YES ▯ DFW allowed, DR forwarded, delivered to destination.
Broadcast/Multicast Trace	Trace broadcast or multicast packets across the segment	Test ARP behavior — confirm ARP is being suppressed by N-VDS rather than flooding the segment
Observation Points	Traceflow shows results at every hop: source vNIC, DFW check, DR forwarding, GENEVE encap, physical network, dest vNIC	Packet drops at DFW on destination host ▯ which rule number dropped it ▯ instantly identify the blocking rule
Packet Drop Reason	Exact reason for drop: DFW rule, routing loop, no route, TTL expired, physical switch blocked	"DROPPED: DFW rule 1023 — Environment Isolation policy denied connection from Dev to Production"

▯ Real-World Example — Traceflow Debug Session:
Developer reports: "My app VM cannot reach the database." NSX admin opens Traceflow. Source: vm-app-03 (10.10.2.50). Destination: vm-db-01 (10.10.3.10), TCP 1433. Result: "DROPPED at vm-db-01 vNIC by DFW Rule 4521". Admin opens Rule 4521 — finds it blocks all traffic from subnet 10.10.2.0/24 to DB tier. The rule was created too broadly. Fix: narrow the rule to allow vm-app-03 specifically. Problem solved in 5 minutes without any packet captures or physical network troubleshooting.

8.2 NSX Intelligence & Flow Analysis

Feature	What It Does	Value
Traffic Flow Visualization	Visual map of all flows between VMs and Security Groups — shows which VMs are talking to which	Security team can see unexpected connections: "Why is the HR VM talking to the Finance DB?"
Recommended Security Policy	Analyzes observed traffic flows and recommends DFW rules to allow only what is actually needed	New application deployed without security policy — NSX Intelligence observes for 7 days, recommends exact allow rules needed. Admin reviews and clicks "Apply Policy."
Anomaly Detection	ML-based detection of unusual traffic patterns — port scans, data exfiltration, lateral movement	"VM-X made 5,000 connections in 60 seconds to 500 different IPs — potential port scan. Alert generated."
Topology View	Shows complete network topology: Segments, Gateways, T0, T1, physical connections	New team member can understand the entire NSX-T network design in minutes from the visual map

8.3 Common Problems & Solutions

Problem	Likely Cause	Diagnostic Command / Tool	Solution
VMs on same segment cannot communicate	TEP (Tunnel Endpoint) connectivity issue — Geneve tunnels not established between hosts	NSX Manager □ Fabric □ Nodes □ check TEP status. CLI: esxcli network ip interface list (check TEP vmk)	Verify Geneve UDP 6081 is allowed between TEP IPs. Check physical switch MTU ≥ 1700 (Geneve adds ~50 bytes overhead). Verify TEP IP is reachable between hosts.
DFW is blocking traffic unexpectedly	Overlapping or incorrect DFW rules — a higher-priority rule is matching and blocking before the allow rule	Use Traceflow to identify exact rule number dropping traffic. Check DFW rule hit counters.	Reorder rules (higher rule number = processed later). Narrow the blocking rule scope. Add more specific allow rule above the blocking rule.

Problem	Likely Cause	Diagnostic Command / Tool	Solution
T0 BGP not establishing with physical switch	Incorrect BGP config, wrong AS number, physical switch ACL blocking BGP (TCP 179), MTU mismatch	NSX Manager □ Networking □ T0 □ BGP □ check neighbor state. CLI on Edge: show ip bgp neighbors	Verify AS numbers match, physical switch allows TCP 179 from Edge TEP IP, check MTU consistency between Edge uplink and physical switch port.
VM cannot reach internet (NAT not working)	Missing SNAT rule, wrong source address in SNAT rule, routing issue on T0	Check T0 NAT rules in NSX Manager. Use Traceflow from VM to 8.8.8.8. Check T0 route table.	Verify SNAT rule has correct source IP range and translated IP. Ensure T0 has a default route (0.0.0.0/0) pointing to physical router. Check physical router accepts the T0 public IP.
NSX Manager UI not loading	NSX Manager VM crashed, disk full, memory exhaustion, certificate expired	SSH to NSX Manager: check services with "get service manager". Check disk: df -h. Check memory: free -m	Restart NSX Manager service: restart service manager. If disk full: delete old logs. If certificate expired: renew via CLI. Ensure Manager VM has sufficient resources.

8.4 Key CLI Commands

Component	CLI Command	What It Shows
NSX Manager	get managers	Status of all 3 Manager cluster nodes
NSX Manager	get cluster status	Overall cluster health — control plane, management, policy API status
NSX Manager	get services	List of all NSX services and their running state
Edge Node	get logical-routers	All logical routers (T0/T1 DR and SR instances) on this Edge

Component	CLI Command	What It Shows
Edge Node	get bgp neighbor summary	BGP neighbor states — is BGP peering established with physical switches?
Edge Node	get route	Routing table — all routes known to T0 SR on this Edge
ESXi Host (via esxcli)	esxcli network ip interface list	All VMkernel ports including TEP — verify TEP IP is correct
ESXi Host (via esxcli)	esxcli network ip route list	Host routing table — verify TEP can reach other TEPs
ESXi Host	net-vdl2 -l	List all VXLAN/Geneve segments (VNIs) on this host
NSX Manager API	GET /api/v1/transport-nodes	REST API: list all transport nodes and their connectivity status

Chapter 9 — NSX-T with Kubernetes & Containers

Modern applications run in containers (Docker) orchestrated by Kubernetes (K8s). NSX-T integrates natively with Kubernetes to provide network virtualization and security for containerized workloads — just like it does for VMs.

9.1 NCP — NSX Container Plugin

The NSX Container Plugin (NCP) is the integration layer between Kubernetes and NSX-T. It runs as a pod inside K8s and translates Kubernetes networking events into NSX-T configuration automatically.

K8s Event	What NCP Does in NSX-T	Result
New Kubernetes Namespace created	NCP creates a dedicated NSX-T Segment and T1 Gateway for that namespace — full network isolation	Each team/app namespace gets its own isolated virtual network automatically. No manual NSX config needed.
New Pod created	NCP allocates an IP from the namespace Segment IP pool and creates a DFW rule allowing pod-to-pod traffic within the namespace	Pod gets IP instantly, connectivity works immediately, and is already security-policy-compliant
Kubernetes NetworkPolicy applied	NCP translates K8s NetworkPolicy rules into NSX-T DFW rules — exact 1:1 mapping	K8s policy "frontend pods can only talk to backend pods on port 8080" becomes a DFW rule enforced at the kernel level
Kubernetes Service (LoadBalancer type)	NCP creates an NSX-T Load Balancer VIP and server pool for the K8s Service automatically	kubectl apply service type=LoadBalancer ▯ NSX-T LB VIP automatically provisioned. No manual LB configuration.
Namespace deleted	NCP removes all NSX-T objects created for that namespace (Segment, T1, DFW rules, LB)	Clean resource lifecycle — no orphaned NSX-T objects when K8s namespaces are torn down

▯ Real-World Example — K8s + NSX-T in Practice:
Platform team creates a Kubernetes namespace "payments-service". NCP automatically creates NSX-T Segment "ns-payments-service" with VNI 200501, T1 Gateway "t1-payments-service", and DFW rules isolating it from all other namespaces. DevOps deploys payment processing pods — they get IPs from the dedicated segment. The Kubernetes NetworkPolicy "only allow API gateway to reach payment pods on port 8443" is translated to an NSX-T DFW rule. All of this happens in under 30 seconds with zero manual networking work.

9.2 NSX-T with Tanzu (VMware Kubernetes)

Tanzu Component	Role	NSX-T Integration
vSphere with Tanzu (Supervisor Cluster)	Runs Kubernetes directly on vSphere — no separate K8s cluster needed	NSX-T provides networking for all Supervisor Cluster workloads. One NSX-T deployment serves both VM and container workloads.
Tanzu Kubernetes Grid (TKG) Clusters	Full upstream Kubernetes clusters managed by Tanzu	NCP installed in each TKG cluster. Each TKG cluster gets dedicated NSX-T Segments and T1 Gateway.
Antrea (Container Network Interface)	Open-source CNI plugin for K8s — native integration with NSX-T	Antrea leverages NSX-T data plane for high-performance container networking with full DFW micro-segmentation

Chapter 10 — Exam Quick Reference & Key Concepts

10.1 Critical Numbers

Topic	Key Value	Context
Max VNI (Virtual Network ID)	16,777,216 (24-bit)	Overlay segments supported — vs VLAN's 4,094 limit
Geneve UDP Port	6081	MUST be open between all TEP IPs — firewall blocking this breaks everything
NSX Manager Cluster Size	3 nodes	Production minimum — odd number for quorum. Single node = lab only.
NSX Manager RAM (Medium)	24 GB per node	For up to 128 transport nodes
NSX Manager Disk	200 GB per node	Stores NSX config database and logs
Edge Node — Large vCPU	8 vCPUs	Most production deployments use Large Edge
Edge Node — Large RAM	32 GB	Required for Load Balancer and VPN services on same Edge
BGP Port	TCP 179	Must be allowed between Edge uplink IPs and physical switch IPs
MTU for Overlay	MTU ≥ 1700 bytes	Physical switches need jumbo frames — Geneve adds ~50 byte overhead to standard 1500 byte frames
Max T0 Gateway per Manager	~160 T0 Gateways	Supported scale depends on NSX-T version and Manager size
NSX-T DFW Rule Types	6 levels/categories	Ethernet, Emergency, Infrastructure, Environment, App, Default

10.2 Common Exam Traps

Trap	Wrong Assumption	Correct Answer
NSX-T = VMware-only	NSX-T only works with vSphere	NSX-T supports ESXi, KVM, bare-metal, containers, and public cloud (AWS, Azure via multi-site)
Geneve = VXLAN	NSX-T uses VXLAN for all overlay tunnels	NSX-T 2.4+ uses Geneve (port 6081). Older versions used VXLAN (port 4789). Geneve is more flexible.
Management Plane outage = network outage	If NSX Manager goes down, VMs lose connectivity	Data Plane is independent. Existing traffic flows continue uninterrupted when Manager is down. Only new config changes are blocked.
DFW is a separate physical appliance	DFW is a firewall VM or physical box	DFW runs INSIDE the ESXi/KVM kernel — at the vNIC level. No separate hardware. Zero additional latency.
T0 is optional	You can have NSX-T without a Tier-0 Gateway	T0 is required for any north-south connectivity (internet access, physical server access). T1 alone = isolated island.
Edge Node handles all routing	All VM-to-VM routing goes through Edge Node	Distributed Router (DR) handles east-west routing inside the hypervisor. Edge only handles north-south (T0 SR) and stateful services.
Security Groups must be static	Security Groups require manually adding VMs	Dynamic Security Groups automatically include VMs matching tag criteria. VM gets tag = instantly in group = DFW rules apply.
Traceflow sends real packets	Traceflow is a packet sniffer	Traceflow injects SYNTHETIC (fake) test packets that do not affect production traffic. Safe to use on live systems.

10.3 Glossary — Key NSX-T Terms

Term / Acronym	Full Name	One-Line Definition
NSX-T	NSX Transformers	VMware's SDN + security platform — virtualizes networking for VMs, containers, bare-metal, multi-cloud
SDN	Software-Defined Networking	Network functions (switching, routing, firewalling) implemented in software, not dedicated hardware
TZ	Transport Zone	Defines the reach of an NSX-T logical network — which transport nodes can participate
TEP	Tunnel Endpoint	VMkernel port on each transport node that creates and terminates Geneve tunnel packets
N-VDS	NSX Virtual Distributed Switch	NSX-T's virtual switch running in each hypervisor kernel — replaces vDS for NSX-T workloads
VNI	Virtual Network Identifier	24-bit ID assigned to each overlay Segment — like a VLAN ID but with 16 million possible values
DFW	Distributed Firewall	Stateful L4 firewall running in every hypervisor kernel — inspects every VM packet at the vNIC level
DR	Distributed Router	Logical router component running in every hypervisor kernel for fast east-west packet forwarding
SR	Service Router	Logical router component running on Edge Nodes — handles stateful services (NAT, LB, VPN, BGP)
T0	Tier-0 Gateway	The "border router" of NSX-T — connects overlay to physical network via BGP/OSPF on Edge Nodes
T1	Tier-1 Gateway	The "tenant router" — connects Segments to T0. One per application or business unit typically.
NCP	NSX Container Plugin	Kubernetes integration component — automatically creates NSX-T objects for K8s namespaces and pods
ECMP	Equal-Cost Multi-Path	Load balancing across multiple T0 SR paths — provides throughput scaling and redundancy
IDFW	Identity Firewall	DFW extension that applies rules based on Active Directory user identity, not just VM IP address

Term / Acronym	Full Name	One-Line Definition
SNAT	Source Network Address Translation	Translates internal VM source IP to a public IP for outbound internet traffic
DNAT	Destination Network Address Translation	Redirects inbound traffic from a public IP:port to an internal VM IP:port
Traceflow	—	NSX-T diagnostic tool — injects synthetic packets to trace path and identify where/why traffic is dropped
BGP	Border Gateway Protocol	Dynamic routing protocol used between T0 Gateway and physical TOR switches to exchange routes
MTU	Maximum Transmission Unit	Packet size limit — must be ≥ 1700 bytes on physical network for Geneve overlay traffic
KVM	Kernel-based Virtual Machine	Linux hypervisor — NSX-T supports KVM as a transport node alongside vSphere ESXi

You have covered all the core NSX-T concepts. Go build secure, software-defined networks!